

Q) Name:- Craddam Sujam
Roll No:- 2023BCD0027

LAB-2

1) Relational model of the given ER model

1) Student (Student-Id, Student-Name, DOB, Course-Id, Age, Address-city, Address-street, Address-door, Address-pin, Address-state)

<u>Student-Id</u>	Student-Name	DOB	Course-Id	Age	Address-city	Address-street	Address-door
			<u>Id</u>				
						Address-state	Address-pin

2) Hobby (Student-Id, Hobby)

<u>Student-Id</u>	Hobby

3) Course (Course-Id, Course-Name)

<u>Course-Id</u>	Course-Name

4) Lecturer-Teaches (Lecturer-Id, Lecturer-Name, Subject-Id, Student-Id)

<u>Lecturer-Id</u>	Lecturer-Name	<u>Subject-Id</u>	<u>Student-Id</u>

5) Subjects (Lecturer-Id, Subject-Id, Course-Id, Subject-Name)

<u>Lecturer-Id</u>	<u>Subject-Id</u>	Course-Id	Subject-Name

6) Attends (Student-Id, Course-Id)

<u>Student-Id</u>	<u>Course-Id</u>

7) Takes (Lecturer-Id, Course-Id)

<u>Lecturer-Id</u>	<u>Course-Id</u>

2 a)

Cardinality ratios and Participation:-

→ Hostel - Room :- Many to Many

Hostel contains many rooms and room can be in multiple order

Participation:- Fully participation

→ Registration Address :- One to many

In registration contain more than one ^(1:N) Address but one Address can be in one registration (1:1).

Participation:- Registration - fully
Address - partially

→ Registration - Courses :- One to Many

Registration contain many courses (1:N) but one course is registered in registration (1:1).

Participation:- Registration - fully
Course - partially

→ Registration - Gender :- One to One

In registration can be selected one gender. One gender can do one registration.

Participation:- Fully participation.

→ Admin - Admin-log :- One to One

One Admin contain only one Admin-log and ~~vice~~ vice-versa.

Participation:- Fully participated both Admin and Admin-log

→ User - Registration - User log :- One to One
One user contains only one user registration and vice versa.

Participation :- Fully participated.

→ Gender - Hostel : Many to One

One Gender contains many hostels but one-hostel can contain one gender.

Participation :- Hostel - Fully

Gender - Partially